

Prepended 22 September 2026 to the record as previously rendered, which follows unchanged.

Update — 22 September 2026 (D-47 closed). Re-measured on the clean cache with `tools/remesure_config_e.py`: none of the 100 blind titles changed prediction, and config E's figures are unchanged at 63.5% precision, 74.0% coverage and 0 fabrications. D-42 did not affect the Phase 1.5 evaluation.

Correction note

Prepended 21 September 2026 to the record as originally rendered, which follows unchanged.

Correction note — 21 September 2026 (D-42). The needs-review figure in this document, 168 of 430 (39.1%), was measured while a classifier defect was caching failed API batches as abstentions. How many of the 168 were affected cannot be recovered, because the cache does not timestamp its entries, so read the figure as an upper bound. At 442 rows the corrected figure is **128 (29.0%)**. Separately, `src/bakeoff.py` classifies through the same cache, so whether the Phase 1.5 evaluation figures carry padded abstentions is an open question, logged as D-47. A padded record removes an answer, so it would lower coverage and could count as a correct abstention on a no-occupation title; precision when the classifier commits is affected only through which titles remain answered. The document below is left as written. See `docs/d42_failure_record.md` for the cause, the evidence and the fix.

From Clustering to Classification

Methodology of the Phase 1 to Phase 1.5 Transition in the Network Visualiser

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1 Overview

This document records the methodological transition the network visualiser underwent between the end of Phase 1 and the close of Phase 1.5. It is written as a decision log rather than a results summary: the point is not that a final approach was chosen, but *why* two earlier approaches were abandoned on evidence, and what the abandonment revealed about the problem.

The headline conclusion is a single sentence. **Grouping a professional network by occupation is a classification problem constrained by the information content of the input, not a clustering problem constrained by algorithm choice.** Every result below is in service of establishing that claim and acting on it.

The transition was triggered by a simple substitution: swapping the synthetic development dataset (83 generated profiles) for a real LinkedIn connections export (430 people). Nothing about the pipeline's code changed at the moment of substitution. What changed was that the real data

violated an assumption the synthetic data had silently satisfied, and that violation cascaded through every downstream stage.

2 Starting Point: Phase 1 on Synthetic Data

Phase 1 built an unsupervised pipeline validated on 83 synthetic profiles drawn from eleven hand-designed archetypes (finance, data science, several engineering sub-types, and so on). On that data the pipeline performed well across four independent validation lenses:

Stage	Metric	Value
Clustering (HDBSCAN)	Silhouette (cosine, noise excluded)	0.4745
Projection (UMAP)	Trustworthiness (384 to 2 dimensions)	0.9840
Graph	Modularity Q of the HDBSCAN partition	0.5224
Graph	ARI, greedy communities vs HDBSCAN	0.8985

These numbers were real, but they were earned on data engineered to be clusterable. The synthetic generator produced profiles by sampling skills and tools from archetype-specific pools, with controlled overlap and a handful of deliberate outliers. By construction, the data contained dense, well-separated regions for a density-based algorithm to find. The validation confirmed the pipeline could recover structure that had been placed there on purpose.

This is the first methodological lesson, recorded here because it was not obvious in advance: **validation on synthetic data confirms that a method can recover known structure; it says nothing about whether real data contains structure of the same kind.**

3 The Substitution and Its Consequence

The real LinkedIn export contained 443 rows. After dropping 13 contacts with neither a role nor a company (LinkedIn withholds position for some connections), 430 people remained. A single diagnostic reframed the entire project:

- 430 people
- 371 distinct job titles
- 336 titles occurring exactly once (singletons)
- **1.16 people per title**

The synthetic data had many people per archetype. The real data had, in effect, one person per title. In the 384-dimensional embedding space this means every person is very nearly their own neighbourhood: there are no dense regions, because density requires repetition and the real network barely repeats.

This is why the HDBSCAN stage did not merely perform worse on real data — it addressed a structure that was not present.

4 Approach One Abandoned: HDBSCAN Clustering

A factorial hyperparameter sweep was run over `min_cluster_size` $\in \{5, 10, 15, 20, 25\}$ and `min_samples` $\in \{3, 5, 10\}$, with `min_samples` always explicit so each row of the sweep table was

reproducible (the default `None` silently resolves to `min_cluster_size`, which would vary two knobs at once).

The sweep exhibited three diagnostic failures, each more informative than the last.

Coverage and separation were in direct opposition. No configuration achieved both. The two extremes bracket the trade-off:

Configuration	Clusters	Coverage	Silhouette
<code>min_cluster_size=5, min_samples=5</code>	15	30.5%	0.5246
<code>min_cluster_size=15, min_samples=3</code>	2	70.7%	0.1382

The highest silhouette (0.5246) was computed on the 30.5% of the network that fell into clusters; the other 69.5% were discarded as noise. Silhouette evaluated on non-noise points rewards discarding hard cases, so “maximise silhouette” degenerates into “cluster a small clean core and ignore the rest”. Every configuration that included more people destroyed separation.

The primary knob went dead. Above `min_cluster_size=15`, results were byte-identical at every `min_samples` value. The two clusters HDBSCAN found were each already larger than 25, so raising the floor never bit. This is not a tuning surface with an unexplored region; it is a plateau, and the plateau is the answer.

A pre-registered gate failed cleanly. The gate — silhouette ≥ 0.35 at coverage $\geq 50\%$, fixed before the sweep — was cleared by no configuration. The gate was set in advance precisely so the result could not be rationalised after the fact.

A controlled test isolated the cause. Job title text alone (without company) was compared against title-plus-company text; removing the company string measurably changed the cluster geometry (the dead knob briefly revived), but no configuration cleared the coverage bar. The limiting factor was not the feature set but the fundamental sparsity: 371 distinct two-word titles do not form density.

HDBSCAN was abandoned for the primary grouping task. It survives only as a possible Phase 3 experiment for finding sub-structure *within* a large occupation group, over company text — the one place structure orthogonal to the taxonomy might exist.

5 Reframing: Clustering Has No Ground Truth

The abandonment of HDBSCAN forced a conceptual clarification that reshaped the rest of the work.

Silhouette measures geometric separation. It has no notion of correctness — a partition can score well while grouping unrelated people, and nothing in the metric detects the error. The moment the project’s priority became *accuracy on real data*, clustering became the wrong tool, because accuracy presupposes a target and clustering has none.

Stating the requirement plainly: wanting “barrister” not to be grouped with “bartender” is a statement that the correct group for a barrister is known in advance. Once categories are known in advance, the task is **classification, not clustering** — assigning each person to a member of a fixed, external taxonomy.

The taxonomy chosen was O*NET 30.3 (U.S. Department of Labor / ETA, CC BY 4.0), which provides a hierarchy of occupations with associated skills, tools, and tasks, and — critically — a lexicon of roughly 57,000 real-world job titles already mapped to occupation codes. Grouping is performed at the level of the 23 SOC major groups, where the taxonomy is most stable: a measurement on the lexicon showed 93.7% of distinct titles map unambiguously to a single major group, against only 84.7% at the finer detailed-occupation level.

6 Approach Two Abandoned: The Embedding Tier

A tiered classifier was built: an exact lexicon match, a fuzzy match, then an embedding-similarity fallback against occupation descriptions, and finally an explicit abstention when nothing cleared threshold. Abstention was treated as a first-class correct outcome — “Member”, “Volunteer” and “Summer Intern” name no occupation, and a blank result is the right one.

Measured per tier against a hand-labelled evaluation set, the tiers separated sharply:

Tier	Mechanism	Accuracy
Exact	Verbatim lexicon hit	85.7%
Fuzzy	Token similarity to lexicon	75.0%
Embedding	Cosine vs occupation description	40.0%

The embedding tier was net-negative. Adding it raised coverage by ten points but lowered precision from 81.8% to 74.1%, and all of its outputs scored within a narrow band (0.55–0.60 cosine), so no threshold could separate its correct answers from its incorrect ones. Inspection of its errors showed the mechanism: it matched on shared words, not meaning (mapping “Student Ambassador — School of Mathematical Sciences” to a mathematics teacher on the word “Mathematical”). On two-word titles, a sentence-embedding model contributes no world knowledge; it is a string matcher with additional cost. The tier was removed.

Company was tested as an input signal across multiple configurations and hurt precision every time. Company encodes industry, which is a distractor for occupational classification: an accountant at a chemicals firm is an accountant, but the employer string drags the embedding toward chemistry. Company was dropped.

7 Establishing the Ceiling

The remaining classifier — deterministic tiers followed by an LLM tier (Claude, title only) for the abstentions — was evaluated against a blind set of 100 titles, labelled before any classifier output was seen. Six configurations were compared. The two informative endpoints:

Configuration	Precision	Coverage
A: deterministic only	62.5%	40.0%
E: Claude, title only	63.5%	74.0%

No configuration cleared the original 85% precision gate. This prompted the decisive measurement: **how often do two humans, given the same titles and the same labelling rules, agree with each other?**

On the 80 titles both labellers committed to (neither abstaining), agreement was **60.0%**. A second, self-consistency check pointed the same way: a single labeller, revisiting the same title boundaries (Office Support versus Business Operations; Life Science versus Healthcare Support), was internally inconsistent on exactly the titles the classifier also failed on.

This is the central finding of Phase 1.5. **A classifier precision of ~63% is not a failure relative to an 85% target; it is performance at the human ceiling.** The 85% gate was never achievable, because the input — a two-word job title — frequently does not contain enough information to determine an occupation, even for a human expert. “Systems Engineer” is genuinely split between Computer and Engineering; “Co-Founder” between Business Operations and Management. The residual error is irreducible ambiguity, not model deficiency.

8 Consequences for Product Design

Because the ceiling is a property of the input, three design decisions follow directly rather than as preferences.

Coverage-first with visible uncertainty. The chosen configuration (Claude, title only) commits to what it can and abstains otherwise, at zero fabrications on the blind set. On the full 430-person network this yields 261 classified (60.7%) and 169 flagged uncertain (39.3%). The uncertain nodes are surfaced to the user for confirmation rather than hidden or guessed. A blank node is honest; a confidently wrong node that cites an authoritative taxonomy is the worst possible outcome, because the user cannot detect it.

Richer input is the only lever that raises the ceiling. No further prompt or model change moves performance meaningfully, because the information is absent from the title. The identified route — deferred to Phase 2 — is self-enrichment: the user supplies their own profile “About” text, which is consented first-party data and adds the signal titles lack. Automated collection of other people’s profile text remains ruled out on Terms-of-Service and data-protection grounds.

Classification replaces clustering in the live pipeline. The SOC major group becomes the grouping key, and the existing graph and centrality stages were retained unchanged by having the classifier emit the same file schema the clustering stage previously produced. The one addition to that contract is an `is_uncertain` flag carried through to the final node table, which is the concrete data artifact of the visible-uncertainty design.

9 An Incidental Finding: Duplicate Titles and Path Metrics

Integrating real data surfaced a latent bug in the Phase 1 centrality code that synthetic data had masked. Path-based centralities convert edge similarity to distance via $d = 1 - s$. When several people share an identical title, they receive identical embeddings and a cosine similarity of exactly 1.0; floating point occasionally returns a value fractionally above 1.0, producing a negative distance, which causes the shortest-path routine to fail with a negative-weight error. Synthetic profiles never repeated text, so the edge case never arose. The fix is to clip the distance at zero. This is recorded because it exemplifies the phase’s theme: **assumptions that hold silently on generated data fail loudly on real data, and finding them is the purpose of testing on real data.**

10 Summary of the Transition

	Phase 1 (end)	Phase 1.5 (end)
Task framing	Unsupervised clustering	Supervised classification
Grouping mechanism	HDBSCAN	O*NET SOC major group
Validation metric	Silhouette (no ground truth)	Precision vs blind labels
Data	83 synthetic profiles	430 real connections
Governing constraint	Algorithm choice	Information in the input
Handling of hard cases	Forced into noise or a cluster	Explicit, visible abstention

Three approaches were ruled out on evidence: HDBSCAN clustering (no density in a 371-title long tail), the embedding classification tier (40% accuracy, net-negative), and company as a signal (industry is a distractor). The chosen classifier operates at the measured human ceiling, abstains rather than fabricates, and hands a per-node uncertainty flag to the visualisation layer.

The most important sentence to carry into Phase 2 is the reframing itself: the limiting factor was never the method, and no better method exists below the information content of a two-word title. The only way forward is more input.

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